DIPLOMA PROGRAMME PRACTICE · SL · SET A

Chemistry / Paper 2

Question booklet · 50 marks · 90 minutes

Answer every part. Show working, units and suitable precision. Use a calculator and a clean Chemistry data booklet. Use separate answer paper for additional working and diagrams.

QUESTION 1 [10]

A salt that loses water

A 4.990 g sample of $CuSO_4 \cdot xH_2O$ gives 3.192 g of dry $CuSO_4$ after repeated heating to constant mass. Use $M(CuSO_4)=159.6$ and $M(H_2O)=18.02$ g mol^{-1}.

(a) Determine x, showing the mole ratio. **[3]**

(b) Describe how to establish constant mass experimentally, including a precaution specific to this residue. **[2]**

(c) All the dry residue is dissolved and made to 250.0 cm^3. Find the Cu^{2+} concentration and the number of sulfate ions in 10.00 cm^3 of this solution. Use $N_A=6.02\times10^{23}$ mol^{-1}. **[3]**

(d) Explain how splattering a little solid out of the crucible during heating would bias x if it were unnoticed. **[2]**

QUESTION 2 [10]

Selecting a conductor

An engineering sample consists of copper containing 70.0% isotope 63 and 30.0% isotope 65 by number. For this question take their relative isotopic masses as exactly 63 and 65. Copper has atomic number 29.

(a) Calculate the relative atomic mass. State the numbers of protons, neutrons and electrons in $^{65}Cu^{2+}$. **[3]**

(b) Write noble-gas configurations for neutral Cu and Cu^{2+}. **[2]**

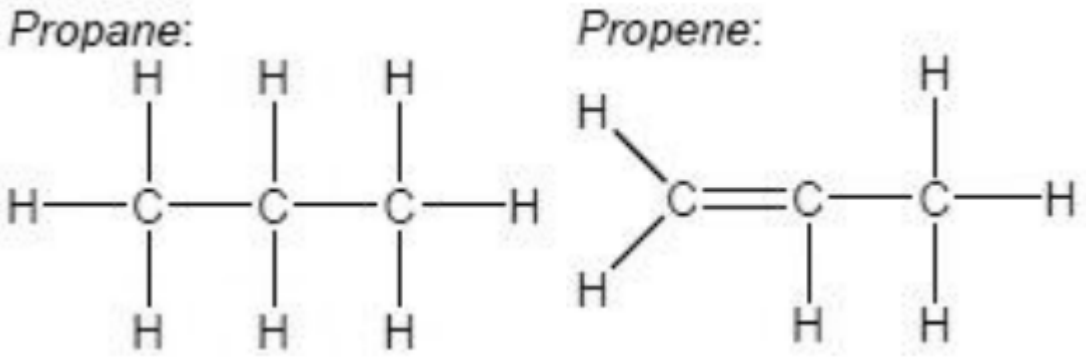
(c) Explain why a copper wire conducts and can be drawn into a thinner wire without losing its metallic bonding. **[3]**

(d) The supplier claims that isotope enrichment will greatly strengthen the wire by disrupting the lattice in the same way as adding a different metal. Evaluate the claim. **[2]**

QUESTION 3 [10]

Two three-carbon feedstocks

The figure shows two candidate feedstocks for making a cable coating. In a separate fuel test, 0.100 mol of the saturated feedstock is burned completely. Use $M(CO_2)=44.0\ g\ mol^{-1}$.



Displayed structures of propane and propene, showing the carbon–carbon double bond in propene.

(a) Identify which feedstock undergoes addition polymerization and draw two consecutive repeat units using a condensed formula with continuation bonds. **[3]**

(b) Give a chemical test distinguishing the two feedstocks. Include the observation for each and state a condition that prevents confusion with a substitution reaction. **[3]**

(c) Write the complete combustion equation for the saturated feedstock and find the mass of CO_2 released in the fuel test. **[2]**

(d) Explain why the polymer can insulate a wire while the metal core conducts. **[2]**

QUESTION 4 [10]

An energy estimate for a fuel cell reactant

Consider the combustion $H_2(g)+\frac{1}{2}O_2(g)\rightarrow H_2O(g)$. Mean bond enthalpies in kJ mol^{-1} are H–H 436, O=O 498 and O–H 463. The enthalpy of condensation of water at the reference temperature is −44.0 kJ mol^{-1}.

(a) Use the bond data to estimate the combustion enthalpy to gaseous water. **[3]**

(b) Calculate the enthalpy to liquid water. Explain why specifying the product state matters. **[2]**

(c) A device converts 55.0% of the estimated energy for liquid-water formation into useful electrical output. Find the output for 4.00 g H_2. Use $M(H_2)=2.016$ g mol^{-1}. **[3]**

(d) Give one limitation of the bond-enthalpy estimate and one reason why the point-of-use CO_2 emission alone cannot establish the fuel's total carbon footprint. **[2]**

QUESTION 5 [10]

Reconciling rate and yield

In a sealed vessel, $2SO_2(g)+O_2(g) \rightleftharpoons 2SO_3(g)$ is exothermic in the forward direction. At one temperature the equilibrium concentrations are 0.200, 0.100 and 0.400 mol dm^{-3} for SO_2, O_2 and SO_3 respectively. At the same temperature, the equilibrium constant for the reversed equation is 0.0250.

(a) Write the equilibrium expression for the forward reaction. Use the reverse-reaction constant to determine the forward constant. **[3]**

(b) The vessel is compressed at constant temperature. Predict the equilibrium change in amount of SO_3 and the change, if any, in Kc. **[2]**

(c) Explain why a catalyst can increase hourly production in a flow reactor without changing the equilibrium yield in a sealed vessel. **[3]**

(d) A manager proposes increasing temperature to maximize both rate and equilibrium yield. Evaluate both parts of the proposal. **[2]**